

Question	Answer	Mark	Mark scheme	Additional guidance
3 (a)	6	P1	for a process to find a relevant area, eg $16 \times 14 \div 2$ (= 112) or $\pi \times \left(\frac{7}{2}\right)^2$ (= 38.4...)	May be implied by $\frac{49}{4}\pi$
		P1	for a process to find the shaded area, eg “112” – “38.4...” (= 73.51...) or “8.96” – “3.07...” (= 5.88...)	
		P1	for a complete process to find the number of bags required for a full or partial area, eg “73.51...” $\div$ 12.5 (= 5.88...) or “112” $\div$ 12.5 (= 8.96) or “38.4...” $\div$ 12.5 (= 3.07...) or [area] $\div$ 12.5 or uses 12.5 in a build up method to exceed [area], eg 12.5×6 (= 75) oe	
	Statement	A1	cao	[area] can be any area but cannot be a length.
		C1	for a valid statement relating to effect on number of bags needed, eg Acceptable examples Will need more bags It will increase Will need an extra bag or will now need 7 bags He won’t have enough There is no change (ft their [area] but must be supported by calculation) Not acceptable examples Will cover less area Needs to change the number of bags needed There is no change (unsupported or incorrect ft their [area]) He may need more bags A calculation using 11 with no supporting statement	
(b)				